SOLITUDE 16MW SOLAR PV

Document Prepared by Voltas Yellow Ltd

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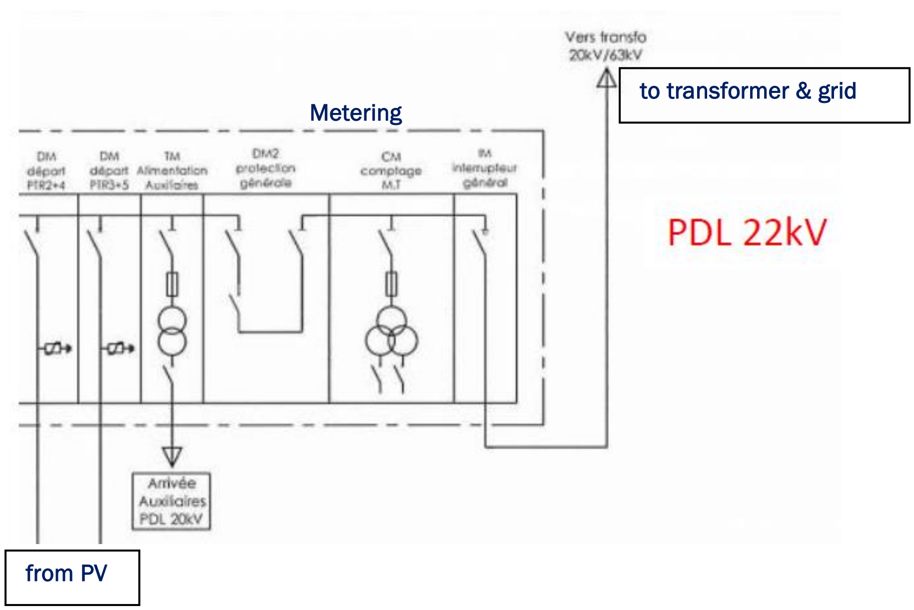
1 PROJECT DETAILS

1.1 Summary Description of the Project

Solitude 16 MW solar PV (“the project”) consists of the construction and operation of a greenfield 16.328 MW_{DC} solar photovoltaic power plant in Solitude near the village of Triolet in the north of Mauritius, in the district of Pamplemousses. It involves the setting up of photovoltaic (PV) panels which will capture solar energy and convey such energy to the convertor station in order to produce electricity exported to the national grid. Electricity in Mauritius is mainly generated from coal and heavy fuel oil1, which is the baseline scenario and can also be considered as the scenario prior to the implementation of the project activity leading to considerable greenhouse gas (GHG) emissions. The project activity undertaken by project promoter Voltas Yellow Ltd will therefore substitute grid electricity by clean and renewable energy and cut down GHG emissions.

The proposed project consists of setting-up an amount of 59 376 solar PV modules with an installed capacity of 16.328 MW to produce electricity which is exported to the grid of the Central Electricity Board (CEB). The CEB in consequence will have to produce less electricity from coal or heavy fuel oil, which are non-renewable sources of energy. The energy production strategy implies a substantial reduction in the production of carbon and its release in the atmosphere, thereby reducing the associated greenhouse impact upon the atmosphere. The figure below provides an overview of the full solar PV system:



A 6.5km 22KV line connects the PV plant to the substation at Riche Terre (Jin Fei). The substation (22/66KV) is the delivery and metering point, including protections and control equipment, then conveying electricity to a 14MVA ABB transformer to rise power to 22/66KV.

The monitoring equipment is composed of two class 0.2 meters for import/export measurement, one of which is used for back-up, located at CEB sub-station.

The project will generate approximately 28,984 tCO_2e emission reductions per year and 202,889 tCO_2e of emission reductions over the 7 years crediting period. During the monitoring period 01-May-2020 to 31-Dec-2021, the project has operated continuously and satisfactorily.

1.2 Sectoral Scope and Project Type

The sectoral scope is Scope 1 – Energy Industries (renewable sources) – ACM0002: Large-scale Consolidated Methodology: Grid-connected electricity generation from renewable sources, Version 20.0. The project is a renewable energy type and is not a grouped project.

1.3 Project Proponent

Organization name	Voltas Yellow Ltd
Contact person	Ramchand COONJUL
Title	Solar O&M Manager
Address	Beau Plan Business Park, North 1A Pamplemousses – Mauritius
Telephone	T : +230 260 5050 / M : +230 58 62 82 15
Email	rcoonjul@greenyellow.fr

1.4 Other Entities Involved in the Project

Organization name	GreenYellow SAS
Role in the Project	Helping for the monitoring report
Contact person	Romane VIEIRA
Title	CSR coordinator
Address	Tour Initiale, GreenYellow, 1, Terrasse Bellini 92800 La Défense - France

Telephone	+33 6 27 52 44 48
Email	rvieira@greenyellow.fr

1.5 Project Start Date

19-December 2018, as the date of commissioning, synchronization and start of GHG emission reductions.

1.6 Project Crediting Period

The project crediting period started on 19-December 2018 and lasts 7 years, until 18-December 2025.

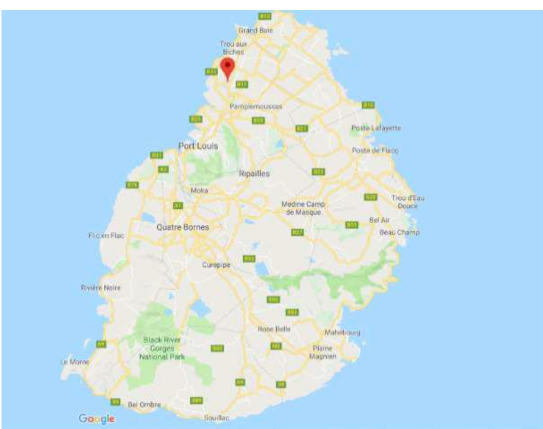
1.7 Project Location

Host Party: Mauritius

Region/State/Province: District of Pamplemousses

City/Town/Community: The site is located in Solitude, which is near the village Triolet.

The project geo-coordinates are: Latitude: 20°04'44"S; Longitude: 57°32'30" E.



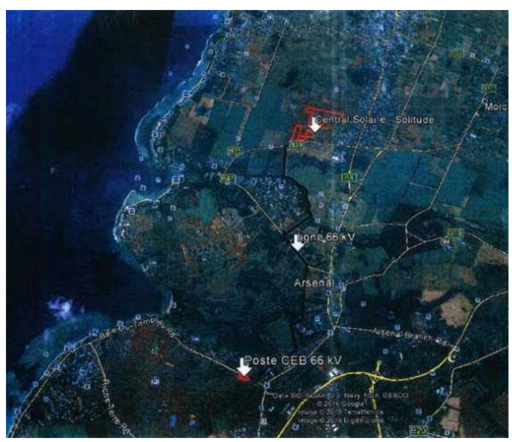


Figure 1: Localisation of project site

1.8 Title and Reference of Methodology

ACM0002: Large-scale Consolidated Methodology: Grid-connected electricity generation from renewable sources, Version 20.0.

1.9 Participation under other GHG Programs

The project is registered under the Clean Development Mechanism[11] (Project #10483) on 31 May 2019, although its GHG emission reduction will either be claimed under the VCS programme or the CDM programme, never both.

1.10 Other Forms of Credit

The project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading.

The project has neither sought nor received another form of GHG-related environmental credits.

1.11 Sustainable Development

The project participants are confident that the proposed project activity will make significant impact on Mauritius sustainable development.

Greenhouse Gas emissions reduction

The project will generate electricity from a renewable and clean energy source, thus avoiding the dispatch of the same amount of electricity produced by fossil-fuelled power plants to Mauritius grid.
Besides, the SOx emissions from coal-fired power plants would also be reduced.

Development of renewable energy

The project will promote and stimulate the commercialisation of grid-connected renewable energy technologies in Mauritius, in line with the Government's renewable energy policy. Adopted in 2011,the vision of Maurice Ile Durable (MID) targeted a 35% share of total energy needs to be fulfilled with renewable energies by 2025. In the Budget Speech 2021/22, the Government of Mauritius announced the Green Energy as a new economic pillar of Mauritius whereby the Renewable Energy (RE) target in the energy mix was increased from 40% to 60% by 2030 together with the announcement of the phasing out of coal in the generation of electricity by the same time.

Energy security

The project will improve energy self-sufficiency of the country which is currently heavily reliant on imported fossil fuels (above 85% according to the governmental Digest of Energy and Water statistics (Ministry of Finance and Economic Empowerment, 2017), alleviating the associated risks of price variations.The government has announced plans to increase use of renewable sources of energy for electricity generation to 60 percent by 2030, and to phase out the use of coal by 2030. It aims to do this through wind farms, solar energy, biomass, wave, and waste-to-energy projects.

Employment opportunities

The project contributes to the local employment throughout its building and operation phases, its workforce consisting of up to 175 workers during the construction phase and approximately 10 to 15 workers in the operations phase. It will also induce indirect employment by increasing the competitiveness of local industry from reducing the country's dependency on fossil fuels. Therefore, Mauritius Government is supportive of the project because the development of solar PV power is in accordance with the national criteria for sustainable development and national policies relating to energy resources and the environment, which will push forward the use of renewable and clean energy across the country.

Technology transfer

This type of renewable energy project will assist building capacities in the country, through advanced technology transfer from industrialized countries. The project will introduce solar PV technology, methods and skills in Mauritius and demonstrate its applicability and efficiency, thus widening its accessibility. The solar panels are manufactured by China Sunergy Co., Ltd., a leading manufacturer of solar cells and modules and closely affiliated with the renowned China Electric Equipment Group (CEEG).

2 SAFEGUARDS

2.1 No Net Harm

An Environmental Impact Assessment (EIA) was required for the solar power plant as a "Power Generating Plant", and it was conducted in line with the ESPA requirements. An EIA license has been obtained on September 5th, 2017 from the Ministry of the Environment and Sustainable Development for the setting-up and operation of the solar PV farm.

We have submitted six EMRs (bi-annual reports) for the operational phase of the Photovoltaic Power Generating Plant of Capacity 16.3 MW, in accordance with section 18(2) (l) of the Environment Protection Act 2002 and clause 4 of the EIA licence ENV/DOE/EIA/1716 granted on 5th September 2017 for the proposed Photovoltaic Plant at Solitude.

The following key environmental and social sensitivities were considered and expected to be of negligible to minor significance in the present phase of the power generating plant:

Landscaping and top soil utilization

Areas that were cleared during construction that are not covered by the PV panels have been covered with top soil in order for vegetation to regenerate. Additionally, to avoid run-off from the water collected by the solar panels, vegetation cover is planted to capture the excess run-off. This vegetation serves to prevent erosion but also provides habitat for fauna species. Land owners are also allowed to cultivate inside the PV farm with compatible crops.

Noise

Limited transient sources of day-time transportation noise and no significant source of noise present during operations as there are no rotating machines at the PV Plant.

Air quality

There are no dust, exhaust or process emissions due to the operation of the PV Plant which affect air quality during operation period. The only source of atmospheric emissions is exhaust gas from vehicles that enter and leave the PV plant.

Water and Irrigation System

All necessary precautions are taken so that the PV development does not impact negatively on the ground water and surface water quality. Necessary measures must be taken to prevent any spillage/leakage of hazardous materials. The existing irrigation network has been left intact, with no modification and the Irrigation Authority is given free access within the perimeter of the PV farm.

Drainage system

The drainage system is also operational and the sceptic tank is emptied as regularly as practical. All drainage infrastructures, including soakaways have been designed and constructed to the satisfaction of the Local Authority, National Development Unit and Road Development Authority. The design of the surface drain network has been carried out such that no storm water within proposed development is channeled towards/into the drain network along the road network of the Road Development Authority.

Biodiversity

No specific biodiversity sensitivity in the Project area and well delineated Project with a limited footprint.

Waste

The project will generate significant quantities of waste during construction and decommissioning phase however these will be easily managed through industrial best practice and the waste management plan specific to the Project.

There is little domestic solid waste generated on site during the operation phase of the project. The waste generated comes from packaging materials such as plastics, cardboards and broken parts. The waste is stored in a suitable container for recycling or disposal to nearest Municipality's waste management facility. The waste transformer oil shall be sent to used-oil recyclers.

In addition to this, green waste is generated through grass cutting. The grasses are left on the ground for composting.

Landscape and visual

The natural landform of the development site is preserved and no activity related to deep excavation including rock quarrying on the plant is carried out.

Traffic

The traffic generated by the project in its current phase is limited in time and intensity as the power generating plant is not located in the busiest part of the village.

2.2 Local Stakeholder Consultation

A Local Stakeholder Consultation (LSC) meeting was held at the site office of Voltas Yellow Ltd on 06/07/2018, with the objective of gathering comments and concerns of stakeholders on the PV Based Power Generator by Voltas Yellow Ltd in Mauritius at Solitude, District of Pamplemousses, Mauritius. The stakeholders were invited by card and by a press advert in the local newspaper. Participants were invited to give their feedback in an open and transparent manner following a presentation on the proposed project activities. The public comments received during the LSC were answered during the 1 hour meeting and no negative comment was received. A fresh LSC was not conducted during the monitoring period as there has not been any change in the project since its operation.

The LSC included:

- the representative of Voltas Yellow Ltd : Mr. Roland de Rosnay;
- the facilitator : Dr. Revin Panray Beeharry;
- some planters/farmers;
- some members of the District Council of Pamplemousses;
- some members of different organisations in the locality.

The consultation meeting began with an informative presentation delivered by the facilitator on Solar PV based power generation and its role in mitigating Global Warming. He explained the phenomenon of Global Warming and the role played by Green House Gases (GHGs) and stressed on the importance of the Kyoto Protocol and the Clean Development Mechanism (CDM) as mechanism for avoiding GHG emission. Dr Revin Panray Beeharry elaborated on the climate change process and elucidated means of combating such a threat by promoting a cleaner technology through the active consideration and the adoption of 'Green Technologies' such as solar PV. Dr Revin Panray Beeharry also stressed on the direct and indirect benefits accrued by such projects and allayed fears of any environmental pollution due to the project activities. Additionally, he explained that the project would add impetus to the socio-economic fabric of the Solitude and Triolet areas and help in sustainable development, as promulgated by the Kyoto Protocol and United Nation Framework Convention on Climate Change.

Stakeholder Consultation Meeting
Attendance Sheet
Date : Friday 06 July 2018
Time: 15.00
Venue: On the site of the project at Solitude opposite the Football Ground

No	Name	Name of Organisation	Designation	Signature	Tel/Mob
1	Bisnoodaye Bissessur	Mon Kali Shakti Mandir	G. Bote	[illegible]	57635403
2	Raghoonauth Soudamanda	Mon Kali Shakti Mandir	P. Bote	[illegible]	~~57284427~~ 58147748.
3	Ramlall Seepaul	Triolet Tae Kwon Do Club	V. President	[illegible]	57282424
4	Bhurtun Devanand	FAREI	SEO	[illegible]	2619216
5	Ibrahim Peeraully	Bilal - Mosque	Sec.	[illegible]	57662250
6	Vijaye Narain	Maheswarnath Institute	Secretary	[illegible]	57545832
7	[illegible]	[illegible] Senior Citizen	Treasurer	[illegible]	59161728
8	Vijaye Chutturdharee	[illegible]	Secretary	[illegible]	55201503
9	Dharam Neegah	~~President MYDS TRIOLET~~	C/President	[illegible]	58508858
10	Prakash Muneeram [illegible]	President [illegible] Triolet Village	Triolet	[illegible]	52526022
11	Abeeluck Iqbal	Secretary Al Madina Society District Councillor Triolet	District C Triolet	[illegible]	52571504
12	D Thumiah	President Vishnu Mandiram	Triolet	[illegible]	58188291
13	V Turnea	Planter			2617886
14	B. GUNGAH HESSOON	Planter	Triolet	[illegible]	2618032

No	Name	Name of organisation	Designation	Signature	Tel/Phone
15	[illegible]	[illegible] Al Muslimah	President	[illegible]	57602109
16	Krisnadath [illegible]			[illegible]	57523210
17	Purgass	District council pamplemousses			
18	Kalichum				
19	Ramen	Chief ~~[illegible]~~ speaker			
20	Ramsing Ramsaw	V.council of Pamplemousses.		[illegible]	52566031
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During the Question and Answer session, it was more clear to the audience as to how the PV project was less polluting and had no side effects compared to conventional ways of power generation. People also exchanged their views on the changes and benefits brought to the village, by the project. The audience looked forward to such good development projects and pondered upon how such projects would fare in the future and whether this project would stand the test of time and last for at least 20 years.

Stakeholders in general expressed their satisfaction at the initiative and welcomed the project. From their perspective, the project activity would accelerate socio-economic development in the area.

Throughout the operation phase of the solar farm, two official environmental reports are prepared on a bi-annual basis which are also transferred to the stakeholders. This tool allows the stakeholders to follow the project's development and raise their queries, if they have any. A log sheet is also available on site whereby visitors and stakeholders are encouraged to register their remarks and grievances, if they have any.

During this monitoring period 01-May-2020 to 31-December-2021, no other local stakeholder comments were received.

Regular exchanges took place with farmers belonging to the cooperative "Les Petits Planteurs". Every year, they come and cut the grass that grows under the solar panels and take it away to feed their livestock.

2.3 AFOLU-Specific Safeguards

Not applicable (this project is not an AFOLU project)

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The commission date of the project activity is 19-December 2018, which is also the start date of GHG emission reductions. During the monitoring period 01-May-2020 to 31-Dec-2021, the project has operated continuously and satisfactorily, generating a total net energy delivered to the grid equal to 39.9 GWh. The only noteworthy event during the monitoring period involved the power phase damages of the inverters which led to under-performance. The root cause of the power phase damage was identified as a manufacturing defect. No changes on the project activity or applicable legislation have occurred during this monitoring period.

3.2 Deviations

3.2.1 Methodology Deviations

No methodology deviations occurred during this monitoring period.

3.2.2 Project Description Deviations

No methodology deviations occurred during this monitoring period.

3.3 Grouped Projects

This is not a grouped project.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO_2e/MWh
Description	Combined margin CO_2emission factor for grid connected power generation in year
Source of data	Representatives of Ministry of Energy of Mauritius
Value applied	0.9915
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO_2e/MWh
Description	Operating Margin CO_2 emission factor for grid connected power generation in year y calculated using the CDM "Methodological Tool to calculate the emission factor for an electricity system".
Source of data	IGES List of Grid Emission Factors
Value applied	1.0282
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO_2e/MWh
Description	Build Margin CO_2emission factor for grid connected power generation in year y calculated using the CDM "Methodological Tool to calculate the emission factor for an electricity system".
Source of data	IGES List of Grid Emission Factors
Value applied	0.8814
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	The percentage share of total installed capacity of solar PV
Data unit	%
Description	The percentage share of total installed capacity of the solar PV in the total installed grid connected power generation capacity in the host country.
Source of data	http://publicutilities.govmu.org/
Value applied	2.6
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Additionality demonstration
Comments	-

Data / Parameter	The total installed capacity of solar PV
Data unit	MW
Description	The total installed capacity of grid-connected solar PV in the host country.
Source of data	http://publicutilities.govmu.org/
Value applied	21.2
Justification of choice of data or description of	-

measurement methods and procedures applied	
Purpose of Data	Additionality demonstration
Comments	-

4.2 Data and Parameters Monitored

Data / Parameter	EGfacility,y
Data unit	MWh/yr
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Measured directly with electricity meter(s) at Interconnection Facilities CEB JINFEI Substation
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	Continuous measurement and at least monthly recording
Value monitored	<table><tr><th>2020</th><th>2021</th><th>Total</th></tr><tr><td>15,893</td><td>24,033</td><td>39,926</td></tr></table>
Monitoring equipment	Electricity outputs are electronically recorded, stored and invoiced monthly to CEB by Voltas Yellow Ltd, based on main meter below (or back-up in case of main meter unavailability or default). The measurement devices of net electrical energy delivered to CEB grid are both located 20/63kV at JINFEI substation and feature the following specifications: <table><tr><td>**Meter designation:**</td><td>Main</td><td>Back-up</td></tr><tr><td>**Meter ownership:**</td><td>CEB</td><td>CEB</td></tr><tr><td>**Make:**</td><td>EDMI</td><td>EDMI</td></tr><tr><td>**Model:**</td><td>MK6E</td><td>MK6E</td></tr><tr><td>**Accuracy class:**</td><td>0.2S (±0.02%)</td><td>0.2S (±0.02%)</td></tr><tr><td>**Serial Number:**</td><td>16538506</td><td>16538507</td></tr><tr><td>**Calibration date:**</td><td>22.06.2020</td><td>22.06.2020</td></tr></table>

QA/QC procedures to be applied	Cross check of measurement results with records for sold electricity. Meter Laboratory of CEB is solely responsible for the selection, installation,calibration, servicing, testing and repairing of all CEB energy meters (CEB Main Meter & CEB Back-up Meter). Voltas Yellow Ltd may install at its own expense a backup electric metering device (Seller Back-up Meter). As per PPA §12.1.4, CEB shall inspect each CEB Meter upon installation and at least once every year thereafter. CEB shall check the certification of CEB Meters through an accuracy test at least once every 4 (four) years thereafter or at any time the readings of Net Energy from the CEB Meter and Seller Back-up Meter differ by an amount greater than 0.5%.
Purpose of the data	Calculation of baseline emissions
Calculation method	-
Comments	As per schedule 12.1.4 of the Energy Supply and Purchase Agreement, the Energy Meters which are installed at Jin Fei Substation have been inspected and load tests have been carried out on the meters to confirm their correctness during the monitoring period.

4.3 Monitoring Plan

The proposed project activity monitoring plan complies with the methodology ACM0002 Grid-connected electricity generation from renewable sources, whereby it is stated that: As per ACM002 provisions for record handling, all data collected as part of monitoring is archived electronically and kept at least for 2 years after the end of the last crediting period.

All measurements are conducted with calibrated measurement equipment according to relevant industry.

Indeed, the quantity of net electricity generation supplied by the project plant to the grid is reliably monitored through calibrated electricity meters and cross-checked with sales records as part of quality assurance/quality control measures on top of CEB diligence. Besides, the reading of the main meter is compared with the check meter at the point of injection.

Data is also daily monitored, analysed and cross-checked by the monitoring team through Qantum (QOS Energy) which is a web based energy management platform that is in real-time communication with the equipment on site.

The data acquisition equipment and installation are subject to regular maintenance and verifications as per the equipment manuals and the meters, to testing as per the ESPA, as stipulated in section 4.2.

Monitoring organization:

The Operations Director of Green Yellow Indian Ocean coordinates and endorses the overall responsibility for all CDM monitoring of the project, including:

- Develop, approve, execute, and improve the CDM Monitoring/Reporting Procedures;
- Organize in-house seminar to inform and train the company staff to the monitoring procedures;

- Ensure that instrumentations and devices are available and properly suited to efficiently perform the monitoring;
- Communicate and coordinate the monitoring work of all business units;
- Validate and electronically archive all monitoring data on a monthly basis throughout the crediting period (and conserve it at least for 2 further years);
- Calculate and report the emission reductions; and
- Coordinate the DOE work during the verification audit.

The Director might appoint a CDM coordinator to delegate him the above specific tasks of monitoring supervision. The Technical/Engineering/Maintenance Department consisting of plant technicians will undertake the technical actions required by the monitoring plan, under the Directors' authority, to collect and record related data.

The Accounting/Sales Department (Chief Financial Officer) will crosscheck, reconcile or consolidate data with multiple sources whenever possible. At minimum, data obtained from the electricity meters is to be crosschecked with the electricity sales receipts. This kind of reconciliation activity will be recorded properly as DOE may request for such information during the verification.

Data collection, consolidation and results analysis is undertaken by a dedicated team adequately trained, well aware of CDM requirements. O&M workflow manual enforced on-site further includes all updated procedures in case of mistake/break-downs, emergency disconnections and cyclone warnings.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The monitoring period for which GHG emission reductions were achieved spans from 01/05/2020 to 31/12/2021. Baseline emissions are calculated according to the methodology as the product of (i) the quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (EGfacility,y = Exports –Imports, in MWh/yr) and (ii) the combined margin CO2emission factor for grid connected power generation in year y (EFgrid,CM,y):

$$\mathrm{BE_y} = \mathrm{EG_{PJ,y}} \times \mathrm{EF_{grid,CM,y}}$$

Where:

$\mathrm{BE_y}$	Baseline emissions in year y (t CO_2/yr)
$\mathrm{EG_{PJ,y}}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)
$\mathrm{EF_{grid,CM,y}}$	Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO_2/MWh)

$EG_{PJ,y} = EG_{facility,y}$ i.e. the quantity of net electricity generation supplied by the project plant to the grid, as monitored and displayed below on a monthly basis.

$\mathrm{EF}_{\mathrm{grid,CM,y}} = 0.9915\ \mathrm{tCO_2/MWh}$, as the Project Proponent have calculated, the DOE validated and the CDM-EB registered.

Year	Imports	Exports	$EG_{facility,y}$	$EF_{grid,CM,y}$	BE_y
	kWh	kWh	MWh	tCO_2/MWh	tCO_2
May-2020	9,756	1,852,174	1,843	0.9915	1,827
June-2020	9,652	1,811,787	1,802	0.9915	1,786
July-2020	9,681	1,627,008	1,618	0.9915	1,604
Aug-2020	10,143	2,109,761	2,099	0.9915	2,081
Sept-2020	9,402	2,134,476	2,125	0.9915	2,106
Oct-2020	8,792	2,115,661	2,107	0.9915	2,089
Nov-2020	8,590	1,919,624	1,911	0.9915	1,894
Dec-2020	9,070	2,397,346	2,388	0.9915	2,367
Jan-2021	9,231	2,309,646	2,300	0.9915	2,280
Feb-2021	8,670	2,183,368	2,175	0.9915	2,156
Mar-2021	9,526	2,328,128	2,319	0.9915	2,299
Apr-2021	9,566	1,771,605	1,762	0.9915	1,747
May-2021	10,160	1,873,277	1,863	0.9915	1,847
June-2021	9,592	1,562,904	1,553	0.9915	1,539
July-2021	10,352	1,839,152	1,829	0.9915	1,813
Aug-2021	10,250	1,932,320	1,922	0.9915	1,905
Sept-2021	7,628	1,607,673	1,600	0.9915	1,586
Oct-2021	9,044	2,169,957	2,160	0.9915	2,141
Nov-2021	8,205	2,206,542	2,198	0.9915	2,179
Dec-2021	8,679	2,360,637	2,352	0.9915	2,332
TOTAL	**185,989**	**40,113,046**	**39,926**		**39,578**

5.2 Project Emissions

No project emissions are expected as the project activity only involves renewable electricity generation from the solar power plant: $PE_y = 0$.

5.3 Leakage

As stated in the applicable methodology, no leakage emissions are considered.

5.4 Net GHG Emission Reductions and Removals

The monitoring period for which GHG emission reductions were achieved spans from 01/05/2020 to 31/12/2021, split by vintage as it follows:

Year	Baseline emissions or removals(tCO2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals(tCO2e)
2020	*15,754*	*0*	*0*	*15,754*
2021	*23,824*	*0*	*0*	*23,824*
Total	***39,578***	***0***	***0***	***39,578***

Ex-ante calculations of Emission Reductions estimated: for the year 2020, the project was supposed to reduce 29,380 tCO_2e. For the year 2021, it was 29,179.

Thus for the monitoring period spanning from 01/05/2020 to 31/12/2021, the project was expected to reduce 8/12*29,380 + 29,179= 48,765 tCO_2e. Ex-post ER is equal to 39,578 tCO2e which is 81% of the expected value. The difference can be explained by overall solar PV lower performance due to the power phase damage which was identified as a manufacturing defect.